An Investigation into Handwriting Speeds: An Analysis of Dominant and Non-Dominant Hands Utilizing Varied Sentence Structures

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Introduction

It is well known that writing with a dominant hand is considerably easier and requires less thought than writing with a non-dominant hand (if a person is not ambidextrous). However, is the speed of a person’s handwriting influenced by the types of words they are writing? Consider for a moment the difference between writing capitalized sentences vs. normal sentences. If the words a person writes are all lower-case, do they need to remove the pencil more often, requiring more writing time? Or can they let the pencil “flow” with no sharp corners or turns? Do writing the straight lines of capital words slow them down? Those are the questions this experiment would like to answer and investigate how sentence structure impacts a person's writing speed in their dominant and non-dominant hand. Handwriting plays an important role in communication, especially in academic writing. It requires dexterity and coordination, and varying sentence complexity can impact handwriting speed. These factors make it very valuable to learn more about the impacts sentence structure can have on handwriting speed.

This report outlines an investigation into the handwriting speeds of different sentence structures. It is broken up into three sections: a Methods section containing how the experiment was conducted in detail, a Results section presenting the findings, and a Conclusion section interpreting the findings and previous analysis. This experiment was sampled from the population of students/staff between 18 and 24 years old at Michigan Technological University in Houghton, Michigan. The samples were collected between April 1, 2024, to April 19, 2024. A total of 138 handwriting samples were collected from 23 different people. The experiment’s observational unit was each individual producing 6 different handwriting samples.

Each individual was asked to write a short sentence consisting of 26 words, and the time it took to completely copy the sentence was recorded as the response variable. The individual was asked to write sentences with either their dominant or non-dominant hand, creating one of the independent variables. Lastly, this was asked for three different sentence structures. Theoretically, writing with a dominant and non-dominant hand should have drastic differences in speed for any given person (which will be later shown in the Results section), and would heavily influence the response variable. Along with this, there should be an impact of handwriting speed between different sentence structures, requiring more time for thought and coordination in hand movements to complete.

The following report will detail the findings and the surprising results that were discovered.

Methods

The following section details the methods in the collection of the data, including the implementation of the data collection, administration of the actual handwriting test, organization and processing of the raw data, and analysis of the collected data.

Data Collection

The experimental procedure was rather straightforward. Comparing how people would write with their dominant hand vs non-dominant hand in upper-case, lower-case, and regular-case. This could be done by creating six different scenarios where the participants would write in each of these situations the same words. Thus, the experiment aimed to design a controlled experiment that met the desired procedure. The experiment necessitated controlling for several variables such as individual writing styles and the person performing in the test. For this reason, a study in which the same person performed multiple parts was chosen. Having each person perform each operation: upper-case, lower-case, and regular-case with both of their hands. The effectiveness of each method of writing was measured using time in seconds as it is very straightforward and without much room for debate. Because of the selection of experiment design, the outside variables can be minimized and only control the desired variables that influence the response variable. After setting the foundation of the experiment, careful steps need to be taken in the selection of individuals for the experiment.

Individuals were selected primarily based on availability. Unfortunately due to a lack of funding, participants couldn't be found based on the proposition of pay but instead had to be found amongst those willing to perform the test for free. This meant that the people who were closest to the experiment designers were the most willing and available subjects. Those individuals were mostly used in the experimental study.

For this reason, the assumption of random sampling is violated. The people were selected based on a convenience sample which would mean that not all members of the population had an equal chance of being selected for the experiment. Especially since the most convenient people would be those closest to the people experimenting which could for some reason due to selection bias in friends also create other selection biases in the people who are sampled. Therefore, the sampling was non-random. Due to the lack of randomization, the generalization of the findings cannot be applied to all students/staff at Michigan Technological University, but rather generalized to individuals similar to the selection criteria stated above.

As previously mentioned in the Introduction, the independent variables were selected based on some of the main ways in which people's handwriting efficiencies may change depending on what they are writing. All proper sentences include upper-case letters and may include more than one. Further, if people use their non-dominant hand would it decrease their ability to efficiently write quickly? Thus both of these variables were controlled for and changed to see if a difference arose, being facilitated by the controlled experiment design.

Lastly, extraneous variables were attempted to be kept out of the model by using the same people to perform each part of the experiment. Instead of worrying about how different people's speeds in writing would affect the data, each person did all six writing prompts, and their times were collected.

Test Administration

After an individual was selected based on the above collection scheme, an ID number was generated for connecting the raw data to a completed Handwriting Study Form (see `MA3740.handwriting.study.form.pdf` attachment). The individual was allowed to read the instructions on the testing procedure, and, if consented to the experiment, their age, gender, major, and dominant hand were recorded. These instructions were made clear to the individual, and they were allowed to ask any questions. The following prompt seen in **Figure 1** below was used for all tests, displayed in either normal, all lower-case, or all upper-case sentence mechanics.

```
Blizzard T. Husky realized it was March 14, and there was
little snowfall here in Houghton at Michigan Technological
University, a departure from the usual winter wonderland.
```

***Figure 1**: The complex sentence used for the testing procedure (upper and lower case). The prompt consisted of 26 words, with a normal capitalization to lower-case ratio of 8:134 (with a total number of letters being 142). The other prompts consisted of the above in all lower-case and all upper-case.*

There were a total of 6 tests administered in a random order. The individual would be asked to copy the prompt while being timed. The time was recorded in seconds (up to the hundredth of a second) using an electronic stopwatch. The information was recorded on paper and then transcribed into a spreadsheet (see

`MA3740.data.collection.wide.csv` attachment) and later imported into R and SAS. Throughout the administration of the test, the individual remained legible, and any response deemed non-legible was removed.

Data Organization

Data was collected from the paper forms and moved to a .csv file in both wide and long format (see `MA3740.data.collection.long.csv` attachment). The data was transposed to long formatting using the R code found in the attached R file. This was done to allow it to be used with R and SAS during the Data Analysis.

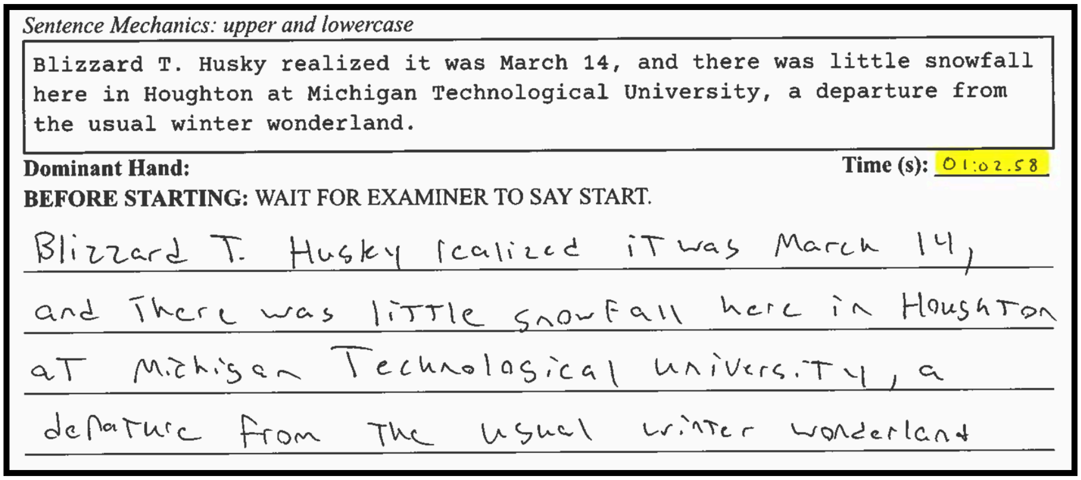
Data Analysis

An analysis of data was performed in both R and SAS, and both files are attached to the following report for reference (see `MA3740.finalproject.R` and `MA3740.finalproject.SAS` attachments). The analysis was performed using a mixed-effect model due to the nature of the way the data was collected. As the subjects would have their own biases in how long it would take them to write the prepared paragraph this variable was controlled for as a random factor in the mixed effects model. Both case and dominant or non-dominant hand usage were the fixed variables as subjects were subjected to each situation. This created a model that would allow for control of individual randomness while still considering how changes in the casing of letters and the hand used would affect the amount of time.

This model was created using the lme4 package in R and the mixed function in SAS. Using the ID of the subject as a random variable and case and dominant or non-dominant hand as another variable. Allowing for interaction between the two variables. Several analyses were then performed on this model in both R and SAS. Getting summaries of each of the effects of each fixed variable and their interaction with the other fixed variable. In R the model was also run through an ANOVA test to check for differences between the populations. In both SAS and R after getting the summaries of the model post hoc comparisons were made between each treatment finding the estimated differences and performing tests on whether or not there is statistical significance to reject the null hypothesis that there is no effect from the fixed effect variables.

Results

After the sample collection, the results were compiled and analyzed through various analyses. **Picture 1** and **Picture 2** showcase an example sample of an individual writing a normal case structure sentence with their dominant and non-dominant hand. Note that **Picture 1** and **Picture 2** come from the same individual. Note the drastic difference in quality and writing time in the yellow highlighted box (see `MA3740.example.1001.pdf` attachment for full form).



Sentence Mechanics: upper and lowercase

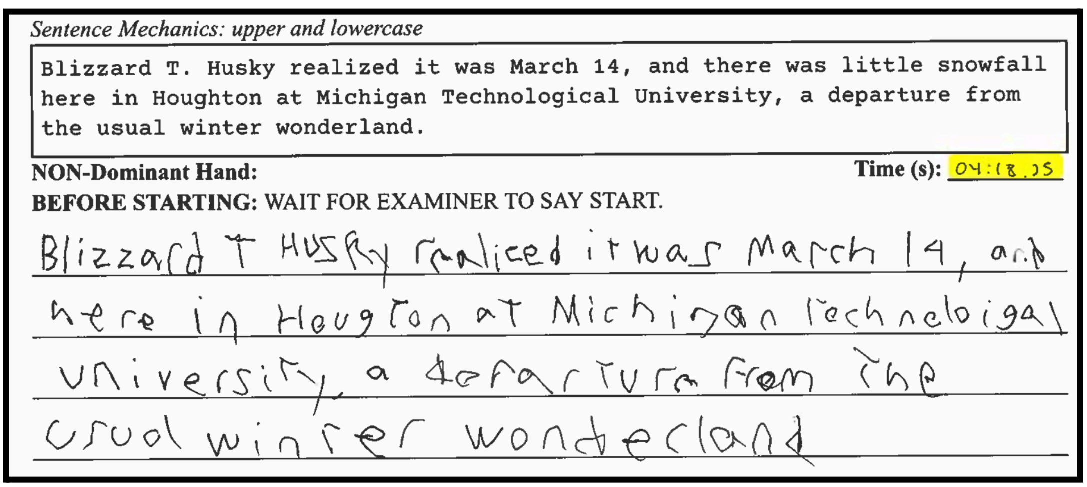
Blizzard T. Husky realized it was March 14, and there was little snowfall here in Houghton at Michigan Technological University, a departure from the usual winter wonderland.

Dominant Hand: **Time (s):** 01:02.58

BEFORE STARTING: WAIT FOR EXAMINER TO SAY START.

Blizzard T. Husky realized it was March 14,
and there was little snowfall here in Houghton
at Michigan Technological university, a
departure from the usual winter wonderland

Picture 1*: An example writing sample from an individual writing with their dominant hand a regular-case structure sentence.*



Sentence Mechanics: upper and lowercase

Blizzard T. Husky realized it was March 14, and there was little snowfall here in Houghton at Michigan Technological University, a departure from the usual winter wonderland.

NON-Dominant Hand: **Time (s):** 04:18.25

BEFORE STARTING: WAIT FOR EXAMINER TO SAY START.

Blizzard T HUSky realiced it was March 14, and
here in Hougton at Michigan Technologal
university, a departura from the
usual winter wonderland

Picture 2*: An example writing sample from an individual writing with their non-dominant hand a regular-case structure sentence.*

An initial analysis of the data using summary statistics and graphics lays the groundwork for the more complex data analysis. Several summary statistics were collected in R and SAS and shown in **Table 1**. Notice the difference in means between dominant and non-dominant hands. There was also an increase in standard deviation when using a non-dominant hand, indicating that our samples varied significantly.

The MEANS Procedure							
Analysis Variable: Time (seconds)							
Hand	Case	N Obs	Mean	Std Dev	Minimum	Maximum	Median
Dominant	Lower	23	59.462	9.665	46.950	85.650	57.170
	Regular	23	60.646	12.219	40.600	97.290	59.510
	Upper	23	105.966	21.887	73.950	159.200	99.410
Non-Dominant	Lower	23	184.175	36.032	134.000	250.700	173.600
	Regular	23	217.352	50.781	122.690	344.900	220.000
	Upper	23	234.110	56.121	156.610	352.530	217.800

Table 1*: Summary statistics of the 6 different scenarios (dominant/non-dominant and upper/lower/regular-case sentence structures. The table was generated using SAS.*

Along with summary statistics in a table format, three different boxplots were created to display the distribution between different sentence structures and their corresponding dominant or non-dominant hand. **Figure 2** showcases the distribution of handwriting for the Dominant Hand in the three different sentence structures using a boxplot graph.

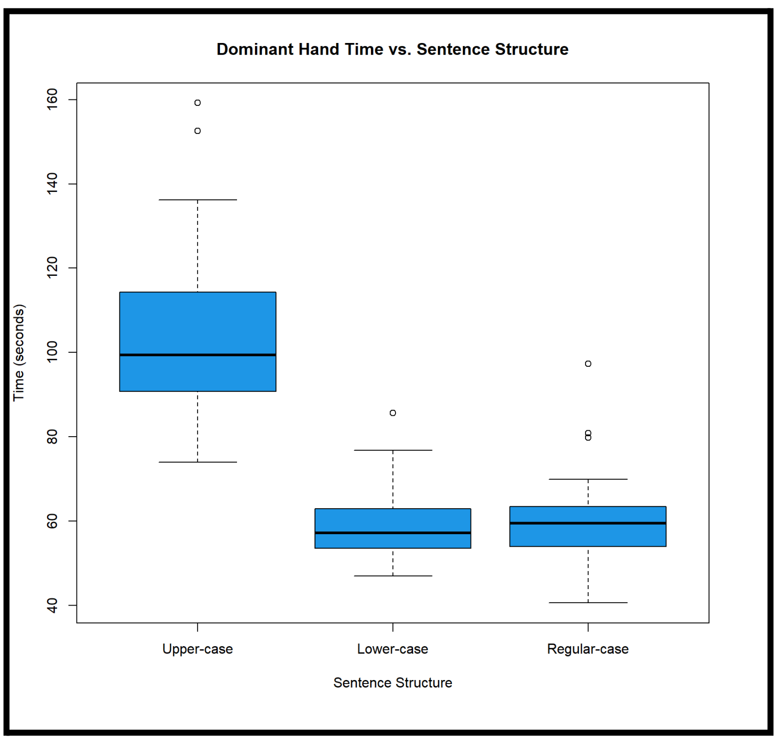


Figure 2*: A box plot depicting the time difference in seconds of the dominant hand for upper-case, lower-case, and regular-case sentence structures. The following was generated using R.*

As seen in **Figure 2**, there is a small spread in terms of lower-case and regular-case, but an overall larger spread and general increase in time for the upper-case when looking at handwriting with the dominant hand. **Figure 3** shows the distribution for the non-dominant hand using a boxplot graph, similar to what was seen in **Figure 2**.

As seen from the two boxplots in **Figure 2** and **Figure 3**, there are drastic differences in the y-axis in terms of scaling. Note that **Figure 2** has a max y-value of 160 seconds while the non-dominant times in **Figure 3** had a minimum y-axis just below 150 seconds. These differences are also shown in **Table 1**, and later shown statistically significant differences in means between the different sentence structures and use of dominant vs. non-dominant hands. The spread between all three is relatively the same, a difference compared to **Figure 2**.

The last boxplot presented in this report is of both dominant and non-dominant hands and the 3 different sentence structures as shown in **Figure 4**. The differences between the dominant and non-dominant hands are more pronounced with all 6 different tests side by side.

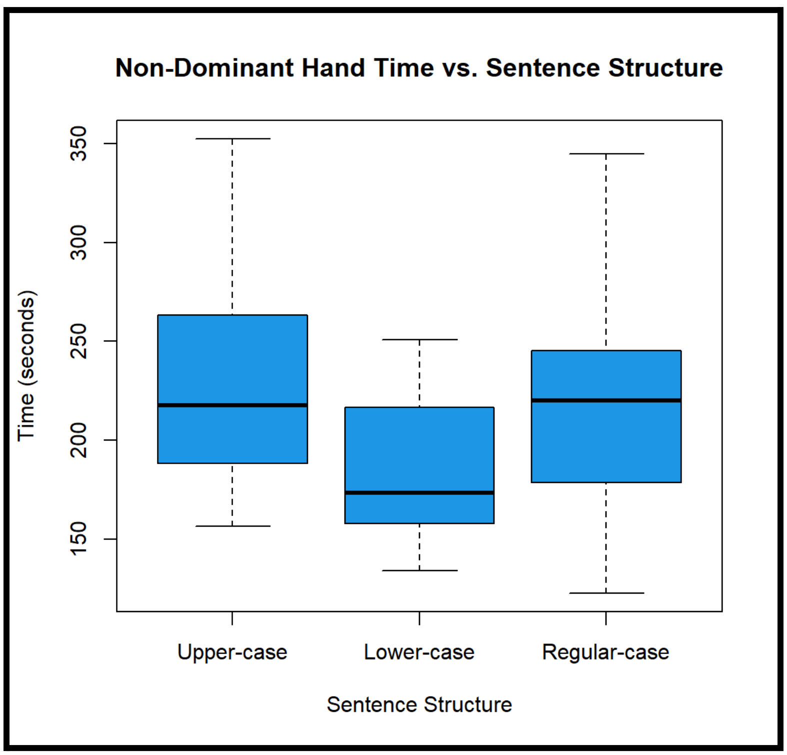


***Figure 3**: A box plot depicting the time difference in seconds of the non-dominant hand for upper-case, lower-case, and regular-case sentence structures.*
The following was generated using R.

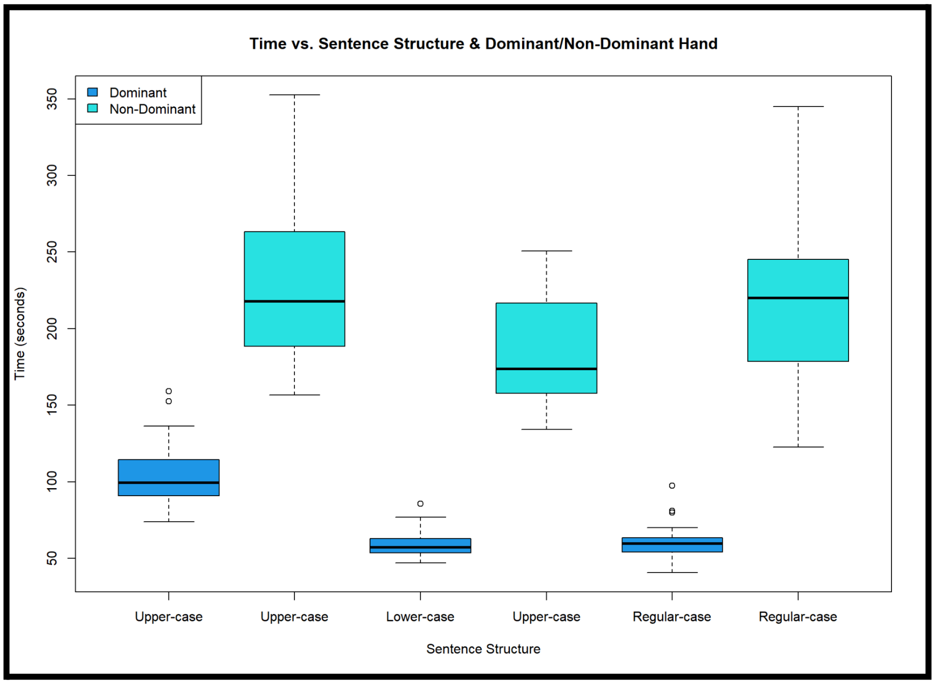


Figure 4*: A box plot depicting the time difference in seconds of the dominant and non-dominant hand for upper-case, lower-case, and regular-case sentence structures. The following was generated using R.*

The variance of each independent variable is noticeable from **Figure 4** to be different and not equal, which will play an important role in the assumptions required for parts of the analysis. An interaction plot was also made to show the relationship between our two independent variables (dominant and sentence structure) with our response variable (time in seconds), which is shown in **Figure 5**. Upper-case and lower-case sentence structures appear to be parallel between dominant and non-dominant, with equal change in time between the two. In terms of regular cases, there is a stronger increase in time for writing between dominant and non-dominant hands. As shown in the previous boxplots, there is a significant difference between the handwriting time when using a dominant hand vs. a non-dominant hand. There was also little difference between lower-case and regular-case when using the dominant hand, as they are starting at roughly the same spot in **Figure 5**.

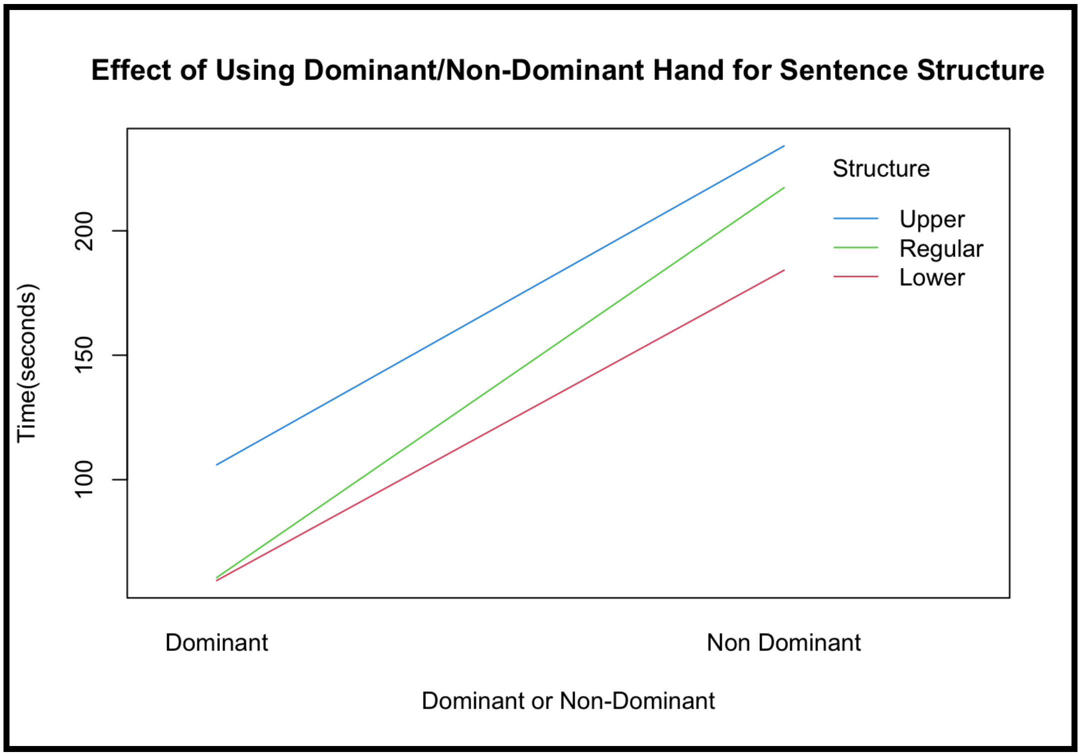


***Figure 5**: An interaction plot where handwriting time in seconds was the response variable and the two categorical (independent) variables were dominant/non-dominant and sentence structure. The following was generated using R.*

The results in **Figure 5** are shown in **Table 2,** which was produced from our model. Note that the Intercept is lowercase and dominant hand. The table also shows that the time increase in seconds from lower-case to regular-case using the dominant hand is almost the same, only being at 1.184 seconds (which is why they start the same in **Figure 5**).

Fixed Effects					
	Estimate	**Std. Error**	**df**	**t-value**	**Pr(>|t|)**
(Intercept)	59.564	7.493	80.401	7.935	1.05E-11
caseRegular	1.184	8.489	110.000	0.139	0.889
caseUpper	46.504	8.489	110.000	5.478	2.75E-07
handNon Dominant	124.713	8.489	110.000	14.691	< 2.00E-16
caseRegular:handNon Dominant	31.993	12.005	110.000	2.665	0.009
caseUpper:handNon Dominant	3.431	12.005	110.000	0.286	0.776

***Table 2**: The summary of Fixed Effects from our Mixed Model. The following was generated using R.*

A Type 3 ANOVA was also performed, and the results can be seen in **Table 3**. The ANOVA shows that there is at least one difference in means between the three different sentence structures, along with the dominant and non-dominant hand. It becomes much more difficult to analyze interactions between case and hand, so no comments will be made at this time for that row in the ANOVA table.

Type III Analysis of Variance Table with Satterthwaite's Method						
	Sum Sq	Mean Sq	NumDF	DemDF	F Value	Pr(<F)
case	54951.000	27475.000	2.000	110.000	33.154	5.39E-12
handNon Dominant	643015.000	643015.000	1.000	110.000	775.899	< 2.2E-16
case:hand	7096.000	3548.000	2.000	110.000	4.281	0.016

***Table 3**: The summary for a Type III Anova using our Mixed Model.*
The following was generated using R.

Along with the ANOVA and Mixed Model, an assessment of normal and equal variance had to be performed. **Figure 6** is a Normal Q-Q Plot of the Residuals from the model, and **Figure 7** is a scatter plot of the Residuals vs. Fitted Values from our Mixed Model.

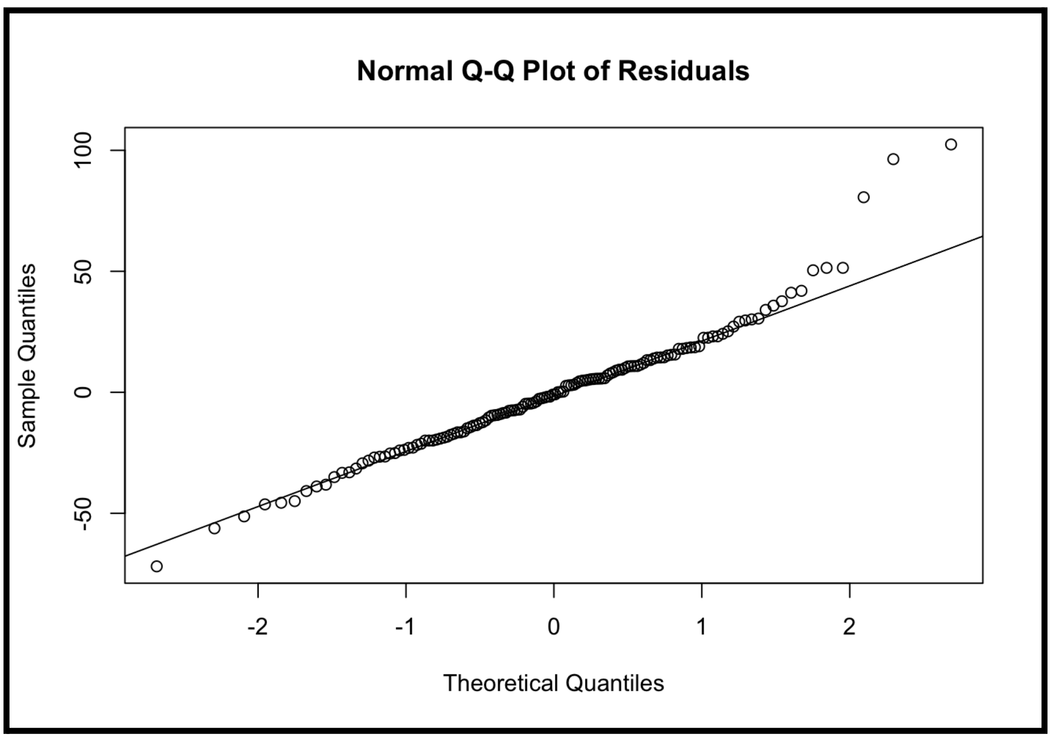


***Figure 6**: A normal Q-Q Plot of Residuals from our Mixed Model.*
The following was generated using R.

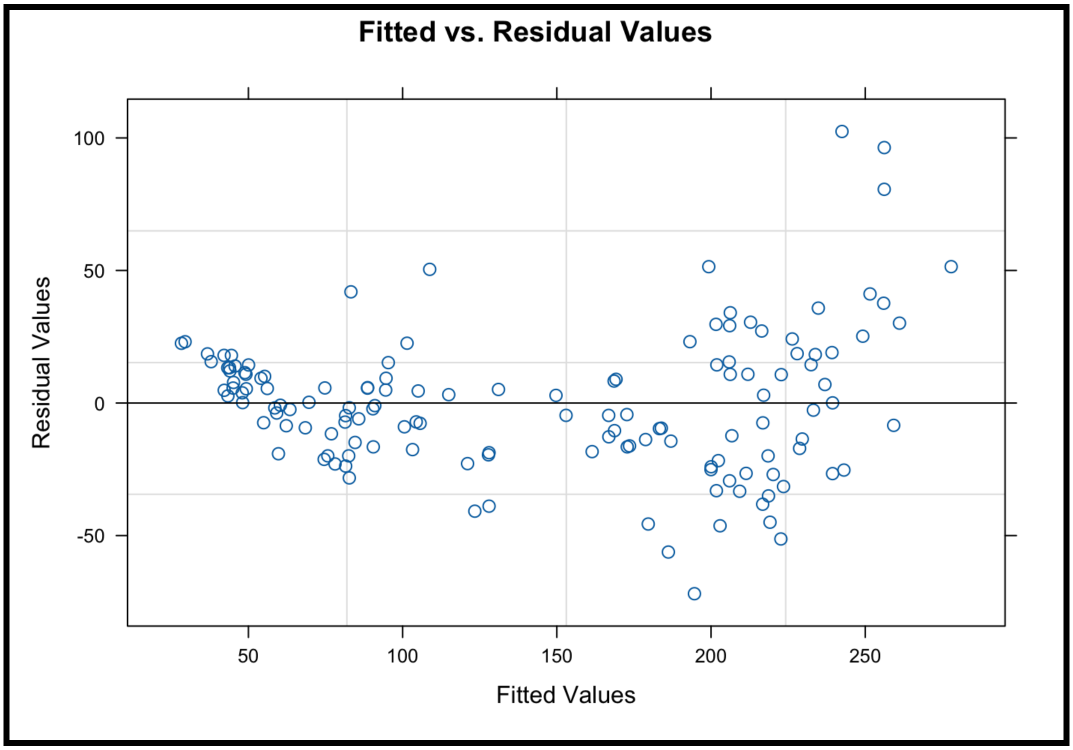


***Figure 7**: A scatter plot of our Fitted vs. Residual Values from our Mixed Model. The following was generated using R.*

When looking at **Figure 6**, the data does appear to follow a roughly linear trend, indicating that our data is normally distributed, even though there were only 23 samples collected. **Figure 7** also indicates that the data collected did not have equal variance, as there appears to be a funnel shape in the scatter plot.

Finally, a post-hoc multiple comparison test was performed on the Mixed Model and the results are shown in **Table 4**. Note that there were observed differences for all but between dominant regular and lower case, which is also seen in the interaction plot in **Figure 5**.

Differences of Least Squares Means										
Effect	hand	case	_hand	_case	Estimate	Standard Error	t Value	Pr > |t|	Lower	Upper
hand*case	Dominant	Lower	Dominant	Regular	-1.184	8.489	-0.14	0.8893	-18.007	15.639
hand*case	Dominant	Lower	Dominant	Upper	-46.504	8.489	-5.48	<.0001	-63.328	-29.681
hand*case	Dominant	Lower	Non Dominant	Lower	-124.710	8.489	-14.69	<.0001	-141.540	-107.890
hand*case	Dominant	Lower	Non Dominant	Regular	-157.890	8.489	-18.60	<.0001	-174.710	-141.070
hand*case	Dominant	Lower	Non Dominant	Upper	-174.650	8.489	-20.57	<.0001	-191.470	-157.830
hand*case	Dominant	Regular	Dominant	Upper	-45.320	8.489	-5.34	<.0001	-62.144	-28.497
hand*case	Dominant	Regular	Non Dominant	Lower	-123.530	8.489	-14.55	<.0001	-140.350	-106.710
hand*case	Dominant	Regular	Non Dominant	Regular	-156.710	8.489	-18.46	<.0001	-173.530	-139.880
hand*case	Dominant	Regular	Non Dominant	Upper	-173.460	8.489	-20.43	<.0001	-190.290	-156.640
hand*case	Dominant	Upper	Non Dominant	Lower	-78.209	8.489	-9.21	<.0001	-95.032	-61.386
hand*case	Dominant	Upper	Non Dominant	Regular	-111.390	8.489	-13.12	<.0001	-128.210	-94.563
hand*case	Dominant	Upper	Non Dominant	Upper	-128.140	8.489	-15.10	<.0001	-144.970	-111.320
hand*case	Non Dominant	Lower	Non Dominant	Regular	-33.177	8.489	-3.91	0.0002	-50.000	-16.354
hand*case	Non Dominant	Lower	Non Dominant	Upper	-49.935	8.489	-5.88	<.0001	-66.759	-33.112
hand*case	Non Dominant	Regular	Non Dominant	Upper	-16.758	8.489	-1.97	0.0509	-33.582	0.065

***Table 4**: A summary of the Post hoc analysis from our Mixed Model.*
The following was generated using SAS.

Conclusion

The whole of the experiment leads to weaker conclusions due to the violations of assumptions discussed previously in the Results and Method sections. While the assumption of normality of the data is met as can be seen in **Figure 6**, there exists violations of both the assumption of homoscedasticity as well as random sampling. The violation of homoscedasticity can be seen in **Figure 7** and **Figure 4**. The violation of homoscedasticity calls into question the model's ability to accurately predict the actual effect of each treatment such as changing to a non-dominant hand or changing the case. The violation of the assumption of random sampling makes generalizations to a greater population more difficult, thus limiting it to populations of the experimenter's close associates.

The data, however, and model still create confidence in being able to suggest that there exist differences between the populations. Both in consideration of the incredibly low p-values in the summary of the linear regression model as shown in **Table 2**. As well as the performed ANOVA test to test if there exists a difference between the populations as shown in **Table 3**. All of this does create confidence in saying that there exists some difference between the populations regardless of the violation of homogeneity as it is only suggesting there exists a difference. The lack of normality would also make it difficult to generalize the results, however, even though it does exist as a convenience sample the people who were sampled are rather representative of the population of Michigan Tech as a whole and the accounting for the random variation between people using the mixed model helps to isolate the effect on people.

It is also noted that the only difference that was not significant was the difference between the population of lower-case writing and upper-case writing. While considering both dominant and non-dominant together and apart using **Table 2** and **Table 4**. Every other group had significant differences using post hoc analysis as can be seen in **Table 4**. Speculatively this could be suggested because regular sentences are mostly lower-case and, therefore, the sentences would be very similar to sentences that are only lower-case.

With better and more sampling it may be possible to draw further conclusions about the actual populations and the more specific differences. Further studies that may solve the problems of this experiment would include a more thorough sampling scheme, as well as selecting a model that doesn't require equal variance.

Also, a continued exploration of the effects of the case as well as using dominant or non-dominant would likely seek to quantify the effects. Such as by using variable text as well as the previously mentioned change in methodology to prevent violations of assumptions.

References

PsychED. (2022, August 4). *Mixed effects models: A conceptual overview using R*. YouTube. https://www.youtube.com/watch?v=VdAxM5SvU8E